

Schrödinger Eq: $\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle$

If we consider $\left\{ \begin{array}{l} E : \text{eigenvalue of } \hat{H} \\ \psi_E : \text{eigenstate of } \hat{H} \end{array} \right\}$

$$\Rightarrow \hbar \partial_t (\psi_E) = E (\psi_E) \quad \text{constant}$$

$$\hookrightarrow \partial_t (\psi_E) = \frac{E}{\hbar} \psi_E$$

$$\Rightarrow |\psi_E(t)\rangle = e^{\frac{E}{\hbar}t} \underbrace{|\psi_E(0)\rangle}_{\text{Initial state, Eigenstate of } \hat{H}}$$

Time-evolution operator.

Eigen basis $\{|\psi_{E_n}\rangle\}$ with $\{E_n\}$
 $\hookrightarrow$ orthonormal basis

$$|\psi(0)\rangle = \sum_n \underbrace{a_n}_{\text{coefficient}} \underbrace{|\psi_{E_n}(0)\rangle}_{\text{Eigenstate of } \hat{H}}$$

$$\Rightarrow |\psi(t)\rangle = \sum_n a_n \underbrace{|\psi_{E_n}(t)\rangle}_{\text{Eigenstate of } \hat{H}} = \sum_n \underbrace{a_n}_{\text{coefficient}} e^{\frac{E_n}{\hbar}t} |\psi_{E_n}(0)\rangle$$

$\uparrow$
 (Schrödinger Eq is linear)

(Q) : How to find $\{|\psi_{E_n}\rangle\}$ in algebraic way?

$$\hat{H}(\hat{p}, \hat{x}) = \underbrace{\frac{\hat{p}^2}{2m}}_{\text{kinetic } E} + \underbrace{V(\hat{x})}_{\text{potential } E}.$$

$$V(\hat{x}) = \frac{k}{2} \hat{x}^2 \quad (\text{Harmonic oscillator})$$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{k}{2} \hat{x}^2 \quad \rightarrow \quad \boxed{\omega = \sqrt{\frac{k}{m}}}$$

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$$

$$= \frac{\omega}{2} \left(\frac{\hat{p}^2}{m\omega} + m\omega \hat{x}^2 \right)$$

$$A^2 + B^2 \stackrel{?}{=} (A+BT)(A-BT) + T \boxed{BA-AB} \quad [BA]$$

only for commutative.

$$\begin{cases} \hat{\alpha}^* = \frac{1}{\sqrt{2\hbar}} \left(\sqrt{m\omega} \hat{x} - \frac{i}{\sqrt{m\omega}} \hat{p} \right) \\ \hat{\alpha} = \frac{1}{\sqrt{2\hbar}} \left(\sqrt{m\omega} \hat{x} + \frac{i}{\sqrt{m\omega}} \hat{p} \right) \end{cases}$$

$$\Rightarrow \hat{H} = \frac{\omega}{2} \left((\sqrt{2\hbar} \hat{\alpha}^*) (\sqrt{2\hbar} \hat{\alpha}) + i[\hat{p}, \hat{x}] \right)$$

$$= \hbar\omega \left(\hat{\alpha}^* \hat{\alpha} + \frac{1}{2} \right)$$

$$\hat{N} := \hat{\alpha}^* \hat{\alpha}$$

Number-operator

Remark • From the def. of $\hat{a}, \hat{a}^*$, we have

$$\begin{aligned} [\hat{a}, \hat{a}^*] &= \frac{1}{2\hbar} \left[\sqrt{m\omega} \hat{x} + \frac{i}{\sqrt{m\omega}} \hat{p}, \sqrt{m\omega} \hat{x} - \frac{i}{\sqrt{m\omega}} \hat{p} \right] \\ &= \frac{1}{2\hbar} (i [\hat{p}, \hat{x}] - i [\hat{x}, \hat{p}]) \\ &= \frac{1}{2\hbar} \cdot 2i [\hat{p}, \hat{x}] = \boxed{1} \end{aligned}$$

Relation between $\hat{N}, \hat{a}, \hat{a}^*$.

$$[\hat{x}, \hat{p}] = i\hbar$$

$$* \boxed{[\hat{a}, \hat{a}^*] = 1}$$

$$\bullet \hat{N} = \hat{a}^* \hat{a} \Rightarrow \hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)$$

$$\bullet [\hat{N}, \hat{a}] = (\hat{a}^* \hat{a} \hat{a} - \hat{a} (\hat{a}^* \hat{a})) = [\hat{a}^*, \hat{a}] \hat{a} = -\hat{a}$$

$$\bullet [\hat{N}, \hat{a}^*] = \hat{a}^*$$

• Since $\boxed{\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)}$ $\hat{H}, \hat{N}$ share their eigen states & eigen values.

$\Rightarrow |\psi_n\rangle$ be an eigen state of $\hat{N}$
with eigen value $\boxed{n}$

$$\hat{N}|\psi_n\rangle = n|\psi_n\rangle$$

$$\hat{a}, \hat{a}^* \rightarrow \text{operators on } \mathcal{H}.$$

$$\mathcal{H} \longrightarrow \mathcal{H}.$$

$$\begin{aligned}\hat{N}(\hat{a}|\psi_n\rangle) &= \hat{N}\hat{a}|\psi_n\rangle \\ &= (\hat{a}\hat{N} + [\hat{N}, \hat{a}])|\psi_n\rangle \\ &\quad \underbrace{\hspace{1cm}}_{\rightarrow -\hat{a}} \quad \text{scalar} \\ &= \hat{a}(\hat{N} - 1)|\psi_n\rangle = \hat{a}(n-1)|\psi_n\rangle \\ &= (n-1)(\hat{a}|\psi_n\rangle)\end{aligned}$$

$\Rightarrow \hat{a}|\psi_n\rangle$ is also an eigen state of $\hat{N}$ with eigen value " $n-1$ "

$$\begin{aligned}\hat{N}(\hat{a}^*|\psi_n\rangle) &= \hat{N}\hat{a}^*|\psi_n\rangle \quad \boxed{+\hat{a}^*} \\ &= (\hat{a}^*\hat{N} + [\hat{N}, \hat{a}^*])|\psi_n\rangle \\ &= \hat{a}^*(\hat{N} + 1)|\psi_n\rangle \\ &= (n+1)(\hat{a}^*|\psi_n\rangle)\end{aligned}$$

$\Rightarrow \hat{a}^*|\psi_n\rangle$ is also an eigen state of $\hat{N}$ with eigen value " $n+1$ "

$$\begin{array}{ccccc}\hat{a}|\psi_n\rangle & \xleftarrow{\hat{a}} & |\psi_n\rangle & \xrightarrow{\hat{a}^*} & \hat{a}^*|\psi_n\rangle \\ \parallel & & & & \parallel \\ |\psi_{n-1}\rangle & & & & |\psi_{n+1}\rangle \\ \textcircled{n-1} & & \textcircled{n} & & \textcircled{n+1}\end{array}$$

$$\begin{cases} \hat{a}|\psi_n\rangle = d|\psi_{n-1}\rangle \\ \hat{a}^*|\psi_n\rangle = c|\psi_{n+1}\rangle \end{cases} \rightarrow \boxed{\text{We'll find c, d later...}}$$

Recall that $\hat{H} = \hbar\omega(N + \frac{1}{2})$

$$\left(\frac{\hat{p}^2}{2m} + \frac{k}{2}\hat{x}^2 \right) \geq 0$$

Energy should be positive.

$\Rightarrow \hat{H}$ has a lower bound.

$\Rightarrow \hat{N}$ has a lower bound

If $n = -2$

$$\Rightarrow \hat{H} \rightarrow \boxed{-\frac{3\hbar\omega}{2}}$$

$$n \geq -1/2$$

For any eigenstate $|\psi_n\rangle$, $\exists$ sufficiently large N

with $\boxed{(\hat{a})^N |\psi_n\rangle = 0}$

$$|\psi_n\rangle \rightarrow |\psi_{n-1}\rangle \rightarrow |\psi_{n-2}\rangle \rightarrow \dots \rightarrow \underline{|\psi_{n-N}\rangle} \rightarrow \dots$$

$\Rightarrow \exists |\psi_{ba}\rangle$: bounded state with

$$\hat{a}|\psi_{ba}\rangle = 0$$

$$(\hat{a})^N |\psi_n\rangle = 0$$

we can choose smallest N s.t.

$$\rightarrow \frac{|\psi_{ba}\rangle}{\sqrt{2}} = C(\hat{a})^{N-1} |\psi_n\rangle$$

unit state

$\hookrightarrow$ Normalization constant

$$(\langle \psi_{ba} | \psi_{ba} \rangle = 1)$$

$\hookrightarrow$ Solve this ODE

Since $\hat{a} = \frac{1}{2\hbar} (\sqrt{m\omega} \hat{x} + \frac{i}{\sqrt{m\omega}} \hat{p})$

$\hat{a} |\psi_{ba}\rangle = 0 \implies \hat{x} |\psi_{ba}\rangle + \frac{i}{m\omega} \hat{p} |\psi_{ba}\rangle = 0$

* $\langle x | \psi_{ba} \rangle = f_{ba}(x)$

$\hat{p} = \int dy |y\rangle \frac{\hbar}{i} \partial_y \langle y|$

$\implies x f_{ba}(x) + \frac{i}{m\omega} \frac{\hbar}{i} \partial_x f_{ba}(x)$

* $\int dx |x\rangle \langle x| = I_H$

$$0 = \int dx |x\rangle \langle x| (\hat{x} |\psi_{ba}\rangle + \frac{i}{m\omega} \int dy |y\rangle \frac{\hbar}{i} \partial_y \langle y| |\psi_{ba}\rangle)$$

$$\stackrel{\text{zero state}}{=} \int dx \left(x |x\rangle \underbrace{\langle x | \psi_{ba} \rangle}_{f_{ba}(x)} + \frac{i}{m\omega} \int dy |x\rangle \underbrace{\langle x | y \rangle}_{\delta(x-y)} \frac{\hbar}{i} \underbrace{\partial_y \langle y | \psi_{ba} \rangle}_{\partial_y f_{ba}(y)} \right)$$

$$= \int dx |x\rangle \left(x f_{ba}(x) + \frac{\hbar}{m\omega} \partial_x f_{ba}(x) \right) = 0$$

$\implies x f_{ba}(x) + \frac{\hbar}{m\omega} \partial_x f_{ba}(x) = 0$ (Simple ODE form)

$\hat{a} |\psi_{ba}\rangle = 0$

$f_{ba}(x) = \text{const.} \times \exp\left(-\frac{m\omega}{2\hbar} x^2\right)$

Bounded soln of the Schrödinger Eq with

$V(x) = \frac{k}{2} x^2$

From $f_{ba}(x)$, we have

$$\langle x | \hat{N} | \psi_{ba} \rangle$$

(Since $|\psi_{ba}\rangle$ is eigen state of $\hat{N}$, we want to find eigen value of $|\psi_{ba}\rangle$)

$$= \langle x | \hat{a}^\dagger \hat{a} | \psi_{ba} \rangle$$

$$[\hat{x}, \hat{p}] = i\hbar$$

$$= \langle x | \left(\frac{1}{2m\omega} (m\omega \hat{a}^2 + \frac{1}{m\omega} \hat{p}^2 + i[\hat{x}, \hat{p}]) \right) | \psi_{ba} \rangle$$

$$= \langle x | \left(\frac{1}{2m\omega} (m\omega \hat{a}^2 + \frac{1}{m\omega} \hat{p}^2 - \hbar) \right) | \psi_{ba} \rangle$$

$$= \frac{m\omega}{2\hbar} x^2 f_{ba}(x) - \frac{\hbar}{2m\omega} \partial_x^2 f_{ba}(x) - \frac{1}{2} f_{ba}(x)$$

$$= 0$$

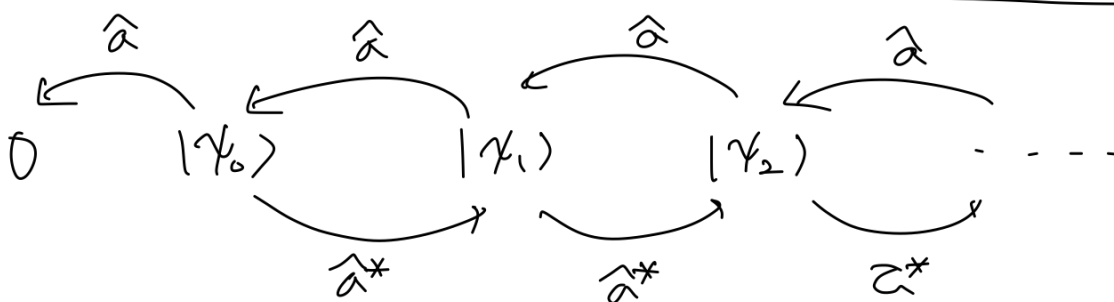
$$\text{Since } \hat{N} |\psi_{ba}\rangle = n |\psi_{ba}\rangle$$

$$\Rightarrow \langle x | \hat{N} | \psi_{ba} \rangle = n \langle x | \psi_{ba} \rangle = 0 \Rightarrow n = 0$$

Non zero

* $|\psi_{ba}\rangle$ is an eigen state with eigenvalue $n=0$

$$\Rightarrow |\psi_{ba}\rangle = |\psi_0\rangle$$



$\hat{a}, \hat{a}^* \Rightarrow$ Ladder operator

$\begin{cases} \rightarrow \text{raising operator } n \rightarrow n+1 \\ \rightarrow \text{lowering operator } n \rightarrow n-1 \end{cases}$

$$|\psi_n\rangle = (\hat{a}^*)^n |\psi_0\rangle \times \boxed{\text{Normalization Const.}}$$

$$\hat{a}^* |\psi_n\rangle = c_n |\psi_{n+1}\rangle$$

$$\hat{a} |\psi_n\rangle = d_n |\psi_{n-1}\rangle$$

$$\Rightarrow |c_n|^2 = \langle \psi_{n+1} | \underbrace{c_n^* c_n}_{\hat{a}^* \hat{a}} | \psi_{n+1} \rangle = (c_n |\psi_{n+1}\rangle)^* (c_n |\psi_{n+1}\rangle)$$

$$= (\hat{a}^* |\psi_n\rangle)^* (\hat{a}^* |\psi_n\rangle)$$

$$= \langle \psi_n | \hat{a} \hat{a}^* | \psi_n \rangle = \langle \psi_n | \hat{a}^* \hat{a} + [\hat{a}, \hat{a}^*] | \psi_n \rangle$$

$$= \langle \psi_n | (\hat{N} + 1) | \psi_n \rangle = (n+1) \langle \psi_n | \psi_n \rangle$$
$$= n+1.$$

$$c_n = \sqrt{n+1} \cdot \cancel{e^{i\phi_n}} \quad \text{not important term}$$

$$\Rightarrow \boxed{\hat{a}^* |\psi_n\rangle = \sqrt{n+1} |\psi_{n+1}\rangle}$$

$$\Rightarrow |d_n|^2 = \boxed{\dots} = n \rightarrow d_n = \sqrt{n}$$

Similar calculation

$$\boxed{\hat{a} |\psi_n\rangle = \sqrt{n} |\psi_{n-1}\rangle}$$

$$\boxed{\hat{a} |\psi_0\rangle = \sqrt{0} |\psi_{-1}\rangle = 0}$$

$$\begin{cases} |\psi_n\rangle = (\hat{a}^\dagger)^n |\psi_0\rangle \times \text{Norm. Const.} \\ \hat{a}^\dagger |\psi_n\rangle = \sqrt{n+1} |\psi_{n+1}\rangle \end{cases}$$

$$\Rightarrow (\hat{a}^\dagger)^n |\psi_0\rangle = (\hat{a}^\dagger)^{n-1} \hat{a}^\dagger |\psi_0\rangle = \dots = \sqrt{n!} |\psi_n\rangle$$

$$\Rightarrow \boxed{|\psi_n\rangle = \frac{1}{\sqrt{n!}} (\hat{a}^\dagger)^n |\psi_0\rangle}$$

$$\left\{ \frac{1}{n!} (\hat{a}^\dagger)^n |\psi_0\rangle : n = 0, 1, 2, \dots \right\} \quad (\hat{N})$$

With Eigenvalue $\{n = 0, 1, 2, \dots\}$.

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) \Rightarrow \boxed{E_n = \hbar\omega \left(n + \frac{1}{2} \right)}$$

Eigenvalue of Hamiltonian operator. $\nwarrow$

with $n = 0, 1, 2, \dots$

Text book.

2.3. Physical Notions: States and Observables.

2.3.1 Pure States.

Postulate 1 An "observable": self-adjoint

operator on $\mathcal{H}$.

(e.g.1) $\hat{x} = \int dx \, x |x\rangle \langle x|$ Matrix representation of $\hat{x}$.

position operator

$$\hat{x}^* = \int dy \, (x |y\rangle \langle y|)^* = \int dy \, y |y\rangle \langle y| = \hat{x}$$

$\Rightarrow \hat{x}$ is self-adjoint & $\hat{x} \in \mathcal{H}_S$ a kind of observable.

(e.g.2) $\hat{p} = \int dx \, |x\rangle \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \langle x|$ is self adjoint.

(p.f) $|\psi\rangle, |\varphi\rangle \in \mathcal{H}$.

$$\langle \psi | \hat{p} | \varphi \rangle = \int dx \, \left(\langle \psi | x \rangle \frac{\hbar}{i} \frac{\partial}{\partial x} \langle x | \varphi \rangle \right)$$

Scalar

$$= \left(\int dy \, \langle y | \psi \rangle \left(-\frac{\hbar}{i} \frac{\partial}{\partial y} \right) \langle \varphi | y \rangle \right) \quad \text{complex conjugate}$$

$$= \langle y | \psi \rangle \left(-\frac{\hbar}{i} \right) \langle \varphi | y \rangle \Big|_{-\infty}^{\infty} + \int dy \, \left(\frac{\hbar}{i} \frac{\partial}{\partial y} \right) \langle y | \psi \rangle \langle \varphi | y \rangle$$

$$= 0 + \int dy \langle \phi | y \rangle \left(\frac{\hbar}{i} \frac{\partial}{\partial y} \right) \langle y | \psi \rangle$$

↑

$$\left(\psi, \phi \rightarrow 0 \text{ as } y \rightarrow \pm \infty \right) = \overline{\langle \phi | \hat{p} | \psi \rangle} = \langle \psi | \hat{p}^* | \phi \rangle$$

$$\Rightarrow \hat{p} = \hat{p}^* \text{ (Since } |\psi\rangle, |\phi\rangle \text{)}$$

↳ self-adjoint & observable.

eg3 Non-observable operators,

$$\hat{x}^* = \hat{x} \Rightarrow i\hat{x} \neq \text{observable}$$

$$i\hat{p} \neq.$$

Def $\langle A \rangle_\psi := \langle \psi | A | \psi \rangle$

$\Rightarrow$ Expectation value of the observable A is the

pure state $|\psi\rangle$



eg.1. $\langle \hat{x} \rangle_\psi = \langle \psi | \hat{x} | \psi \rangle$ $\hat{x} = \int dy \, y |y\rangle \langle y|$

$$= \int dy \langle \psi | y \rangle \otimes \langle y | \psi \rangle$$

$$= \int dy \, y \underbrace{|\langle \psi | y \rangle|^2}_{\text{probability density}}$$

eg.2 $\langle \hat{p} \rangle_\psi = \langle \psi | \hat{p} | \psi \rangle$

$\left(\frac{\hbar}{i} \partial_x \right) = \int dx \underbrace{\langle \psi | x \rangle} \frac{\hbar}{i} \partial_x \underbrace{\langle x | \psi \rangle}$

$= \boxed{????}$

From Schrödinger Eq,

$$i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle \Rightarrow \partial_t |\psi\rangle = \frac{\hat{H}}{i\hbar} |\psi\rangle$$

$$\Rightarrow \partial_t \langle \psi | \hat{x} | \psi \rangle$$

$$= (\partial_t \langle \psi |) \hat{x} | \psi \rangle + \langle \psi | \hat{x} | (\partial_t |\psi\rangle)$$

$$= \left(-\frac{1}{i\hbar} \langle \psi | \hat{H} \right) \hat{x} | \psi \rangle + \langle \psi | \hat{x} \left(\frac{\hat{H}}{i\hbar} | \psi \rangle \right)$$

$$= -\frac{1}{i\hbar} \langle \psi | (\hat{H} \hat{x} - \hat{x} \hat{H}) | \psi \rangle \quad \hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$= -\frac{1}{i\hbar} \langle \psi | [\hat{H}, \hat{x}] | \psi \rangle$$

$$= -\frac{1}{i\hbar} \langle \psi | \left[\frac{\hat{p}^2}{2m} + V(\hat{x}), \hat{x} \right] | \psi \rangle$$

commute.

$[\frac{\hat{p}^2}{2m} + V(\hat{x}), \hat{x}]$

$[\hat{p}^2, \hat{x}] = \hat{p} [\hat{p}, \hat{x}] + [\hat{p}, \hat{x}] \hat{p}$

$= -2i\hbar \hat{p}$

$$= -\frac{1}{i\hbar} \langle \psi | -\frac{2i\hbar}{2m} \hat{p} | \psi \rangle$$

$$= \frac{1}{m} \langle \psi | \hat{p} | \psi \rangle$$

$$\frac{d}{dt} \langle \hat{x} \rangle_\psi = \frac{1}{m} \langle \hat{p} \rangle_\psi$$

$$\Rightarrow \frac{d}{dt} \langle \hat{x} \rangle_\psi = \frac{1}{m} \langle \hat{p} \rangle_\psi \quad \rightarrow \text{momentum operator.}$$

$$\hookrightarrow \dot{x} = \frac{p}{m} \quad \leftarrow p = mv$$

Postulate 2. • We can only measure the eigenvalue of $\hat{A}$ from the experiment. (projection operator defined as $|\lambda\rangle\langle\lambda|$)

• $P_\psi(\lambda) = \|P_\lambda|\psi\rangle\|^2 \leftarrow \lambda: \text{one of the eigen value of } A$

$\uparrow$ $\hookrightarrow$ probability that we measure λ from the state ψ .
 $\langle\psi|\psi\rangle = 1$

In harmonic Oscillator, we obtained eigen values of

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{k}{2} x^2 \quad \text{give as } \boxed{E_n = \hbar\omega(n + \frac{1}{2})} \quad \star$$

$\{0 \leq P_\psi(\lambda) = \|P_\lambda|\psi\rangle\|^2 \Rightarrow \{P_\psi(\lambda) : \lambda \in \sigma(A)\}$ is a kind of probability distribution.

$$\sum_\lambda P_\psi(\lambda) = \sum_\lambda \|P_\lambda|\psi\rangle\|^2 = \sum_\lambda \langle\psi|P_\lambda^* P_\lambda|\psi\rangle$$

$$= \sum_\lambda \langle\psi|\lambda\rangle\langle\lambda|\lambda\rangle\langle\lambda|\psi\rangle$$

$$\stackrel{\mathbb{I}_H}{=} \sum_\lambda \langle\psi|\lambda\rangle\langle\lambda|\psi\rangle = \langle\psi|\psi\rangle = 1.$$

Def For $\forall |\psi\rangle \in H$, $\|\psi\|=1$.

$S_\psi := \{ e^{i\alpha} |\psi\rangle \mid \alpha \in \mathbb{R} \}$ is called a ray in H .

with $|\psi\rangle$ as a representative

Proposition $|\psi\rangle \in H, |\phi\rangle \in H$ with $\|\psi\|=\|\phi\|=1$.

$$IP \left(\begin{array}{l} \text{System prepared in state } |\psi\rangle \\ \text{observed in state } |\phi\rangle \end{array} \right) = |\langle \phi | \psi \rangle|^2$$

$$(P1) \quad \hat{A} = |\phi\rangle\langle\phi|$$

Eigen value of $\hat{A} = \begin{cases} 1 \rightarrow e^{i\alpha} |\phi\rangle \\ 0 \rightarrow \text{orthogonal to } |\phi\rangle \end{cases}$

$$LHS (*) = P_\chi(\lambda=1) = \|\hat{A}|\psi\rangle\|^2$$

$$= \langle \psi | \phi \rangle \langle \phi | \phi \rangle \langle \phi | \psi \rangle = |\langle \psi | \phi \rangle|^2$$

